Inequality of Opportunity and Income Redistribution

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Opportunities, such as access to education or family background, shape income inequality by influencing the chances of economic success. Unequal opportunities create uncertainty about whether success is merit or luck based. We examine how uncertainty impacts preferences for redistribution, comparing it to more transparent forms of luck. Using a US representative sample, we elicit redistribution decisions in two environments: when workers' earnings are influenced by luck directly ("lucky outcomes") or indirectly ("lucky opportunities"). Spectators redistribute less and are less sensitive to differences in luck under opportunities. Inferential challenges drive this gap, which implies that individuals are less meritocratic than previous findings suggest.

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I. Introduction

The design of redistribution policies—from progressive taxation to social welfare programs—depends on what citizens consider to be a fair distribution of income. Experimental work has found that most people hold meritocratic fairness attitudes: They tolerate income disparities that are merit based but choose to redistribute when income differences result from circumstances beyond individuals' control, such as luck (e.g., Cappelen et al. 2007, 2013; Almås, Cappelen, and Tungodden 2020). However, the widespread acceptance of meritocratic principles contrasts sharply with recent trends in income inequality in the United States, where support for redistribution has remained stagnant despite a rising impact of circumstances beyond people's control (Chetty et al. 2014; Ashok, Kuziemko, and Washington 2015; Bozio et al. 2024).

We propose that this apparent disconnect between experimental findings and observed redistribution trends is partly due to limitations in how prior studies implemented luck in lab settings. Previous work has primarily focused on redistribution behavior in situations where luck arises in a simple and transparent way, making it easy to assess whether inequality is merit or luck based. By contrast, in a survey we conducted of US households, we found that Americans identify unequal opportunities that merely affect someone's chances of success as the most critical source of luck in life.[1] For example, the main sources of luck identified by these US households include the family one is born into and the education one receives. The impact of these opportunities on an individual's success is hard to assess without also observing that individual's effort. Thus, a fundamental feature of unequal opportunities is that they create uncertainty about the

O'Donoghue, Ryan Oprea, Matthew Rabin, and seminar participants at the briq Institute on Behavior and Inequality, the Economic and Social Research Institute, the Verein für Socialpolitik Annual Conference 2022, the 2022 Big Data in Economics at ZBW–Leibniz Information Centre for Economics, the 2022 North American Economic Science Association Conference, the 2023 CESifo Area Conference on Public Economics, the Organizations Research Group seminar at Ludwig-Maximilians University Munich, and the 2024 Spring Behavioral and Experimental Economics Research Conference. The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of New York or the Federal Reserve System. The experiment reported in this paper was preregistered in the American Economic Association Randomized Controlled Trial Registry in October 2021, under the ID AEARCTR-0008147 (https://www.socialscienceregistry.org/trials/8147). Details on our experimental and survey items are available in the supplementary materials (https://joyzwu.github.io/papers/prsw_supplementary.pdf). The experiment was reviewed and granted an exemption by the Institutional Review Board at Cornell University. This work has been supported in part by a grant from the Cornell Center for Social Sciences. This paper was edited by Anna Dreber Almenberg.

[1] Approximately 1,000 panelists from the New York Fed's *Survey of Consumer Expectations* participated in this survey. We provide a detailed description and analysis of the survey in app. sec. B.1 (apps. A and B are available online).

reasons for someone's success. This uncertainty creates inferential challenges that prior studies have not examined.

To illustrate the difficulty of implementing meritocratic fairness ideals when there are unequal opportunities, consider whether the success of Microsoft founder Bill Gates was due to merit or luck. On one hand, he was fortunate to attend one of the few high schools that offered unlimited access to a computer programming terminal. This is just one of many lucky opportunities in Gates's lifetime that were instrumental to his eventual success (Frank 2016). However, he is also known for a fierce work ethic, famously stating that he "didn't believe in weekends; didn't believe in vacations." Both luck and effort were instrumental in his professional success, and assessing the exact impact of each factor poses a challenging inference problem.

We recruited 2,400 workers on an online labor market to perform a real-effort task and randomly paired them to compete for a fixed prize in a winner-take-all environment. Then, we asked 1,170 individuals ("spectators") from the New York Fed's *Survey of Consumer Expectations*—a nationally representative survey of US households—to choose the final earnings allocation for pairs of workers. We randomly assigned spectators to one of two luck environments. In the *lucky outcomes* environment, we selected the winner of each worker pair either by the number of tasks completed (merit) or by a coin flip (luck). In the *lucky opportunities* environment, the winner of each pair was the worker with the higher number of tasks completed (merit) multiplied by a randomly assigned productivity multiplier (luck). These productivity multipliers capture, in reduced form, the type of advantages that create uncertainty about the source of success: circumstances beyond individual control such as family background, educational quality, and early-life conditions. We generate exogenous variation in the probability that inequality is merit based—that is, the probability that the winner of a pair is the worker who completed more tasks—by varying the coin-flip probability (in lucky outcomes) or the relative productivity multipliers (in lucky opportunities). We use this probability to compare redistribution across the different luck environments while holding luck (the likelihood that the winner of a pair is the worker who performed better) constant.

In our baseline treatment, spectators are not informed about the probability that inequality is merit based. Instead, spectators observe information about the environment that conveys the importance of merit and luck in determining the winner. The inferential challenge they face is using this information to infer how likely it is that the inequality between workers is merit based. Under *lucky outcomes*, spectators observe the probability that the winner was determined by a coin flip. This feature of the lucky outcomes environment is linearly related to the probability that inequality is merit based. Under *lucky opportunities*, they observe the productivity multiplier of each worker. Mapping this feature of the lucky opportunities

environment to the probability that inequality is merit based is more challenging since a small change in the relative advantage of the winner can imply large changes in the probability that the winner is the worker who completed more tasks.[2]

Our main result is that spectators redistribute less when inequality arises from lucky opportunities compared to lucky outcomes. On average, spectators redistributed 27.6% of earnings from winner to loser under lucky outcomes but only 23.4% under lucky opportunities. This redistribution gap emerges only when there is uncertainty about whether inequality is merit based. Specifically, when performance alone determines the winner (i.e., under equal productivity multipliers or zero probability of a coin flip), spectators redistribute similarly across environments. As the likelihood of luck-based inequality increases, redistribution responds strongly under lucky outcomes but remains more muted under lucky opportunities, creating a redistribution gap between environments. We estimate that a 10 percentage point increase in the probability of luck-based inequality leads to a 3.7 percentage point increase in redistribution under lucky outcomes but only a 2.0 percentage point increase under lucky opportunities.

We introduce two additional treatments to investigate the mechanisms underlying the differences in redistribution across luck environments. First, to further examine the role of inferential challenges, we provide direct information about the likelihood that inequality is merit based. Consistent with spectators facing inferential challenges, we find that this information significantly reduces the redistribution gap between luck environments. We show that the information acts by changing the way in which spectators incorporate unequal opportunities into their redistribution decisions. Second, to isolate the role of potential effort responses in driving our results, we implement an "ex post" lucky opportunities condition in which workers learn about their multiplier only after completing the task. We find no significant difference in redistribution between the baseline and ex post conditions, indicating that effort responses do not explain the redistribution gap.

We primarily contribute to a broad literature that studies how the source of inequality affects redistribution. Numerous experimental studies have demonstrated that people tolerate more inequality when earnings differences stem from differences in effort or individual choices, rather than luck (Cappelen, Sørensen, and Tungodden 2010; Cappelen et al. 2013, 2022, 2023; Durante, Putterman, and van der Weele 2014; Mollerstrom,

[2] The inferential challenges spectators face are not unique to our experimental design, but an inherent feature of many environments in which workers have unequal opportunities. Specifically, the probability that the outcome is merit based is decreasing and convex in the winning worker's relative advantage. In app. secs. B.2 and B.3, we show that the same relationship prevails even if lucky opportunities are not multiplicative.

Reme, and Sørensen 2015; Almås, Cappelen, and Tungodden 2020).[3] We show that whether luck arises through unequal opportunities or unequal outcomes significantly affects how much inequality individuals tolerate. We also advance the methodology of redistribution experiments by introducing a common scale for quantifying the importance of luck under uncertainty. This scale provides a portable definition of luck that can be widely applied to different tournament environments.[4]

Our work also contributes to the emerging literature examining how individuals make redistribution decisions when luck stems from differences in workers' circumstances (Andre 2025; Dong, Huang, and Lien 2025; Bhattacharya and Mollerstrom, forthcoming). Andre (2025) investigates whether spectators hold workers responsible for the unequal incentives they face. He finds that disparities in randomly assigned piece-rate wages produce large differences in worker performance. However, spectators reward workers according to their effort and irrespective of how differences in piece rates impact workers' performance. Dong, Huang, and Lien (2025) examine redistribution when luck arises through different employment offers or training quality. They find that spectators only partially account for these differences in advantages when making redistribution decisions. Similarly, Bhattacharya and Mollerstrom (forthcoming) examine redistribution decisions when luck determines whether individuals can work at all. They find that spectators tolerate more inequality when luck determines who is allowed to work than when luck directly determines outcomes.

Our paper advances this literature in two fundamental ways. First, we study an environment where the impact of unequal opportunities is uncertain, mirroring real-world settings where the relationship between advantages and success is ambiguous. This design reveals that inferential challenges—the difficulty in assessing how much luck versus merit drives inequality—are key to understanding redistribution decisions. Second, while prior work focuses on how spectators reward performance differences that arise from unequal opportunities, we examine situations where equally hard-working individuals can achieve vastly different outcomes purely because of their circumstances. This shows that spectators have different attitudes toward redistribution even when there are no performance differences across workers.

[3] Other research on the factors influencing redistribution has examined the role of other-regarding preferences (e.g., Charness and Rabin 2002), fairness ideals (e.g., Konow 2000; Cappelen et al. 2007), and perceptions regarding mobility and income inequality (e.g., Cruces, Perez-Truglia, and Tetaz 2013; Kuziemko et al. 2015; Alesina, Stantcheva, and Teso 2018; Fehr, Müller, and Preuss 2024).

[4] Our approach builds on Cappelen et al. (2022), who study redistribution under lucky outcomes using variation in the probability that inequality is merit based. We extend their framework by deriving and implementing an analogous measure for opportunity-based luck, enabling us to systematically compare how redistribution responds to different forms of luck while holding the magnitude of luck constant.

Finally, we contribute to the literature that examines heuristics and biases in the inference process by documenting their consequences in an important economic context. We document that in the lucky opportunities environment, redistribution is close to linear on the workers' multiplier difference. This "linearization heuristic" leads spectators to underestimate the impact of unequal opportunities and consequently to redistribute less than they would under lucky outcomes. This finding is consistent with previous work showing that individuals often struggle to estimate nonlinear relationships (Larrick and Soll 2008; Levy and Tasoff 2016; Rees-Jones and Taubinsky 2020) and fail to solve even simple Bayesian updating problems (Benjamin 2019), particularly in environments with uncertainty that require contingent reasoning (Martínez-Marquina, Niederle, and Vespa 2019). The results from our information provision treatment are consistent with work showing that drawing attention to an attribute increases individuals' sensitivity to that attribute (Conlon 2026) and that people focus on readily available information (Enke 2020; Graeber 2023).

II. Theoretical Framework

This section presents a stylized model of spectators' redistribution decisions when spectators care about the source of inequality but worker performance is unobservable. Our aim is not to test a specific model of spectator behavior, but rather to use the model to clarify our experimental setup and provide a common framework for quantifying the impact of luck on outcomes, regardless of its source. The model setup closely follows that of Cappelen et al. (2022) and applies the framework to different luck environments.

Consider an impartial spectator who observes initial earnings in a winner-take-all environment in which two randomly paired workers compete for a fixed prize. Spectator i's task is to choose r_i, the fraction of income to redistribute from the winner to the loser. We model r_i as a function of fairness views and beliefs about how likely it is that outcomes are based on merit. Formally, let f_i denote the share of total income for the worker with the lower performance that spectator i deems to be fair, and let $1 - f_i$ denote the fair share for the higher-performance worker. We assume that spectators choose r_i to minimize the quadratic difference between the fair allocation $(f_i, 1 - f_i)$ and the actual allocation $(r_i, 1 - r_i)$:

$$U(r_i, f_i) = -(r_i - f_i)^2. \tag{1}$$

If spectators know with certainty that the winner is the worker who performed better, then they implement the fair allocation, $r_i^* = f_i$. However, in our experiment, as is often the case in reality, spectators do not observe workers' performance. This creates uncertainty about whether the

best performer won or lost. As a result, spectators maximize the expected utility

$$\mathbb{E}(U(r_i, f_i)) = -\pi(r_i - f_i)^2 - (1 - \pi)(r_i - (1 - f_i))^2, \quad (2)$$

where π denotes the probability that the best performer won, and $1 - \pi$ is the probability that the best performer lost. In other words, π represents the probability that the outcome is merit based, while $1 - \pi$ represents the probability that the outcome is luck based. Thus, we often refer to $1 - \pi$ as our measure of the degree of luck in the environment, which we use to compare redistribution choices across different luck conditions in our experiment.

In an interior solution to (2), the optimal level of redistribution is

$$r_i^* = \pi f_i + (1 - \pi)(1 - f_i). \quad (3)$$

Equation (3) highlights that redistribution depends on both the spectator preferences about the fair share of earnings for the lower- and higher-effort worker (f_i) and how likely it is that the outcome was due to luck ($1 - \pi$). Provided the fair share for the lower-performance worker satisfies $f_i < 1/2$, the optimal level of redistribution increases in $1 - \pi$. In other words, the more likely it is that the better-performing worker lost, the more spectators redistribute. When performance solely determines the winner ($1 - \pi = 0$), spectators redistribute the fair share to the loser, $r_i^* = f_i$. When outcomes are due to pure luck ($1 - \pi = 1/2$), spectators equalize earnings and choose $r_i^* = 1/2$.

Outcome luck and opportunity luck.—We consider two work environments with different processes for generating worker earnings. In the lucky outcomes environment, the winner of each worker pair is determined by either luck or performance. Specifically, we assume that with some probability $q \in [0, 1]$, a coin flip determines the winner of the match. Conversely, with probability $1 - q$ the winner is the worker who exerted more effort.

In the lucky opportunities environment, each worker is assigned an individual-specific productivity multiplier, m_i, which influences their earnings. We assume that the distribution of productivity multipliers is exogenous to the distribution of effort, meaning that a worker's productivity is determined by factors outside their control, such as innate abilities or acquired skills (e.g., Mirrlees 1971).[5] The winner of each worker pair is determined by comparing their final output, which is calculated by multiplying each worker's productivity multiplier by their performance level.

We assume that spectators observe the features of the environment that determine whether the outcome is merit based, π, but do not observe

[5] This approach captures the idea that some workers may be more fortunate than others because of their inherent talents or the opportunities they have had to develop their skills, leading to higher earnings even when putting in the same level of effort as their peers.

π directly. In the lucky outcomes environment, spectators observe the coin-flip probability q. To infer π from q, spectators must use Bayesian updating, which implies $\pi = 1 - q/2$. In the lucky opportunities environment, spectators observe the productivity multipliers of each worker. Spectators can use this information to estimate π. Without loss of generality, assume that worker 1 wins the tournament, which means $m_1 e_1 > m_2 e_2$. The probability that the worker who won is the worker who exerted more effort given the information observed by the spectator is

$$\pi = \Pr(e_1 \geq e_2 | m_1 e_1 > m_2 e_2, m_1, m_2). \tag{4}$$

When inferring π from the features of the environment, spectators can encounter cases with or without uncertainty about whether the best performer lost. First, if $m_1 \leq m_2$, then $\pi = 1$ and there is no uncertainty. Intuitively, if worker 1 wins despite being less or equally productive as worker 2, then they must have exerted more effort than worker 2. Second, if $m_1 > m_2$, there is uncertainty and equation (4) becomes

$$\pi = \frac{\Pr(e_1 \geq e_2, m_1, m_2)}{\Pr(m_1 e_1 > m_2 e_2, m_1, m_2)} = \frac{1/2}{\Pr(e_2/e_1 < m_1/m_2)} \geq \frac{1}{2}, \tag{5}$$

where the equality follows under the assumption that the distribution of effort is independent of the multiplier.[6] Expression (5) shows that π depends on the relative multiplier m_1/m_2 or, equivalently, on the relative advantage conferred to worker 1. We show in appendix section B.2 that π is decreasing and convex in this relative advantage for any log-normal distribution of effort. Thus, the probability that the best worker lost $(1 - \pi)$ or, equivalently, the probability that the winner was luck based, is increasing and concave in the winning worker's relative advantage. Intuitively, this concavity arises because the density of e_2/e_1, which measures the relative effort of the disadvantaged worker compared to the advantaged worker, is centered around 1 (equal effort) and decreases for larger values of e_2/e_1. Therefore, even small relative advantages can greatly impact who wins, whereas the incremental effect of increasing this relative advantage matters less.[7]

[6] This assumption is not essential for our analysis but simplifies the exposition of the theory. We also show in sec. V that this assumption is consistent with our experimental data.

[7] The convexity of π (concavity of $1 - \pi$) in the relative advantage is not unique to the way we chose to model unequal opportunities. We show in app. sec. B.3 that implementing opportunity luck as an additive head start also leads to a convex mapping from lucky head starts to the likelihood that the winning worker exerted more effort (π) for the empirical distribution of worker effort that we observe in the experiment. This is because $e_2 - e_1$, which is the relevant measure of relative effort in this case, has the most probability mass near zero (equal effort), and therefore the convexity intuition above still applies (see the last paragraph in app. sec. B.2 for the complete theoretical argument). We also show in app. sec. B.2 that the same intuition readily extends to other common effort distributions.

This framework illustrates how differences in spectators' beliefs about π can lead to differences in redistribution when luck affects workers indirectly through unequal opportunities versus when luck can affect workers' earnings directly. A reason for potential variation in beliefs about π across our luck environments is that estimating π may be more challenging under lucky opportunities, particularly if spectators fail to appreciate that even a small multiplier advantage can correspond to a considerable probability that the winning worker is not the one with the higher performance.

III. Experimental Design

A. Production, Earnings, and Redistribution Stages

The experiment follows the impartial-spectator paradigm in Cappelen et al. (2013). It consists of three stages: a production stage, an earnings stage, and a redistribution stage. In the production stage, workers complete a real-effort task to encrypt three-letter "words" into numerical code (Erkal, Gangadharan, and Nikiforakis 2011). Workers have 5 minutes to correctly encrypt as many words as possible using a dynamic and randomly generated codebook for each word (Benndorf, Rau, and Sölch 2019). In the earnings stage, we randomly pair workers and determine the winner based on some combination of performance and luck. We initially allocate earnings of \$5 to the winner and \$0 to the loser. The exact interaction between luck and performance forms our main experimental treatments, which we describe in section III.B.[8] Workers know that a third party may influence their final earnings, but the spectator's identity is entirely anonymous to them.[9]

In the redistribution stage, spectators choose how much income to redistribute from the winner to the loser. Spectators make 12 redistribution decisions involving different real pairs of workers. To incentivize spectators to respond truthfully, we randomly select and implement one of their 12 decisions. Spectators may redistribute any amount between \$0 and \$5 in \$0.50 increments at no cost.[10] We present each decision as an adjustment

[8] We designed all experimental programs in oTree (Chen, Schonger, and Wickens 2016). Screenshots of our experimental design and procedure are available in the supplementary materials (https://joyzwu.github.io/papers/prsw_supplementary.pdf).

[9] To combat the influence of anchoring effects in the spectators' redistribution choices, we inform spectators that we did not tell workers whether they won or lost, or the exact amount they will earn in each case. Spectators know that we only informed workers that they could earn up to \$5 and that winning against their randomly assigned opponent increases their chances of earning those \$5. This design removes any confounding issues relating to spectators' unwillingness to take earnings away from what workers might expect.

[10] We implemented costless redistribution to preserve comparability with prior experimental work on redistributive behavior (e.g., Cappelen et al. 2013; Almås, Cappelen, and Tungodden 2020). This approach also allows us to isolate how spectators respond to varying sources of luck.

schedule (see fig. A1 [figs. A1–B6 are available online] for an example of a redistribution choice). For example, an adjustment of \$1.50 implies \$3.50 for the winner and \$1.50 for the loser. The first option is always a \$0.00 adjustment, which we label a "no"-adjustment choice. The remaining {\$0.50, ... , \$5.00} redistribution choices are labeled as a "yes"-adjustment choice and denote the final earnings for both the winner and the loser: that is, {(winner gets, loser gets)} = {(\$4.50, \$0.50), (\$4.00, \$1.00), ... , (\$0.50, \$4.50), (\$0.00, \$5.00)}.

B. Spectator Treatments

We implement within-subject variation in how likely it was that the winner was merit based (i.e., π). Spectators always observe features of the environment that are informative about the probability that the inequality between workers is merit based. Moreover, the experiment embeds between-subject variation in the source of luck (lucky opportunities vs. lucky outcomes), the timing of when luck is revealed to the workers (ex post vs. ex ante), the information available to spectators about the importance of luck (full vs. partial), and the timing of workers' awareness about the source of luck (after vs. before). Table A1 (tables A1–B3 are available online) provides an overview of these between-subject treatments.

1. Lucky Outcomes versus Lucky Opportunities

We randomly assign one-third of the spectators to redistribute earnings under lucky outcomes and two-thirds to redistribute earnings under lucky opportunities. In the lucky outcomes condition, we select the winner based on a coin flip with probability q and based on the higher number of correct encryptions with probability $1 - q$. In the lucky opportunities condition, we generate inequality of opportunity by randomly assigning productivity multipliers to workers. For example, a worker with a multiplier of 1.2 who solved 20 encryptions would have a score of 24, while a worker with a multiplier of 3.0 who solved 10 encryptions would have a score of 30. The winner in each pair is the worker who has the higher score. We draw the multiplier for each worker i from the distribution $m_i = 1$ with probability .05, $m_i = 4$ with probability .05, and $m_i \sim U(1, 4)$ with probability .9.[11] We round all multipliers to the nearest tenth.

We do not inform spectators about the workers' actual performance, though we do provide some information about how likely it is that luck determined the winner. In the lucky outcomes condition, spectators know

[11] We chose a relatively high upper bound of $m = 4$ to ensure that it is possible to generate $\pi = 1/2$ in our lucky opportunities treatment ($\pi = 1/2$ when the advantaged worker wins always; as if pure chance determines the winner).

the probability q that we determine the winner by a coin flip, but not whether a coin flip actually determined the winner. Spectators also know that we do not reveal this probability to workers, though workers know that there is some unstated chance that a coin flip determines their outcomes. In the lucky opportunities condition, spectators know each worker's multiplier. We inform spectators that workers only know of their own multiplier and the distribution from which it is drawn, but do not know anything about the worker they compete against. Figure A1 provides an example of a redistribution decision in our lucky outcomes condition and an example of a redistribution decision in our lucky opportunities condition.

2. Varying the Importance of Luck

We implement within-subject variation in how likely it is that the better-performing worker won across worker pairs. We implement this variation in the lucky outcomes environment by changing q across matches. We implement this variation in the lucky opportunities environment by varying the relative workers' multipliers across worker pairs. We control for the importance of luck by introducing a portable common metric across environments: the probability that the winner in a given pair completed more encryptions (π). When $\pi = 0.50$, there was a 50% chance that the winner of the match was the one who performed better—for example, when a coin flip determined the outcome ($q = 1$) in the lucky outcomes environment or when the ratio between worker multipliers is sufficiently large (so that the worker with the high multiplier always won the match) in the lucky opportunities environment. When $\pi = 1$, there was a 100% chance that the match's winner performed better—for example, when $q = 0$ or when the winning worker had a weakly lower productivity multiplier than the losing worker.

Spectators make redistribution decisions for a total of 12 worker pairs. Each worker pair corresponds to a unique value of π drawn from one of the following 12 bins:

$$\pi \in \{\underbrace{\{0.50\}}_{\text{Bin 1}}, \underbrace{\{0.51, \ldots, 0.54\}}_{\text{Bin 2}}, \underbrace{\{0.55, 0.56, \ldots, 0.59\}}_{\text{Bin 3}}, \ldots, \underbrace{\{0.95, 0.96, \ldots, 0.99\}}_{\text{Bin 11}}, \underbrace{\{1\}}_{\text{Bin 12}}\}. \quad (6)$$

For each spectator, we randomly draw one value of π from each of the 12 bins. This ensures that every spectator makes a decision with $\pi = 0.5$, $\pi = 1$, and that the remaining values are evenly distributed throughout the support of π. We present the 12 decisions in random order.

The key information we present on each decision is the multiplier of each worker pair, (m_i, m_j), or the ex ante probability that a coin flip q determined the winner. Every possible pair (m_i, m_j) and coin-flip probability q correspond to a unique π value. The mapping from π to q is given by the formula $q = 2(1 - \pi)$. To map π to a multiplier pair, (m_i, m_j), it is sufficient to

consider the relative multiplier $m \equiv \max\{m_i, m_j\}/\min\{m_i, m_j\}$. Given any relative multiplier m, we examine all possible worker pairs and compute the fraction of times that the winner was the worker who solved more encryptions, which is the empirical counterpart to the probability that the winner is the worker who performed better, π.[12] This method yields a mapping from m to π as depicted in figure A2*B*, which shows that π converges to $1/2$ for all $m > 3$. For a given π, we then select a random worker pairing with a corresponding relative multiplier.

3. Timing of Opportunity Luck

We also randomly vary the timing of when luck occurs. In our baseline lucky opportunities condition, we inform workers of their multipliers before they begin working on the encryption task. In the ex post lucky opportunities condition, workers learn their multipliers only after completing the task. We assign half of the spectators in the lucky opportunities conditions to the baseline treatment and the other half to the ex post treatment.

4. Information Intervention

We randomly assign half of the spectators in each treatment to receive full information about π. In the lucky opportunities condition, we present the following additional text on the redistribution decision screen: "Based on historical data for these multipliers, there is a $[\pi \times 100]\%$ chance that the winner above completed more encryptions than the loser." In the lucky outcomes condition, the equivalent text is as follows: "There is a $[\pi \times 100]\%$ chance that the winner above completed more encryptions than the loser." As noted above, the value of π varies from decision to decision. Figure A1 shows an example of the decision screens for the information treatments for both luck environments.

5. Workers' Awareness about Rules

Finally, we vary the timing of when workers learn how luck plays a role in determining outcomes. In the rules-before condition, we inform workers

[12] With 800 workers per condition, there are

$$\binom{800}{2} = 319{,}600$$

possible pairings. Since we can assign the higher multiplier to either worker, that creates 639,200 observations that we can use to calculate π for each relative multiplier, m. Using this method, we compute, for each m, the fraction of all possible pairings in which the winner completed more encryptions than the loser.

that effort multipliers or a coin flip can influence the outcome before they start the task. In the rules-after condition, we inform workers that multipliers or a coin flip can influence the outcome after they complete the task. We randomly assign half of the spectators in the ex post lucky opportunities and lucky outcomes conditions to the rules-before treatment and half to the rules-after treatment. By construction, we assign all participants in the baseline lucky opportunities condition to the rules-before treatment. Spectators have complete information about when workers learned how we determine the winner.

C. Comprehension Checks and Belief Elicitation

To ensure that spectators understand the design details, we implement several comprehension questions after they see the initial instructions about the worker task. These questions test spectators' understanding of how luck can affect outcomes and their awareness of when workers learn about the importance of luck. Spectators can only continue once they select the correct answer, and we briefly explain why the answer is correct once they submit it. Therefore, these questions serve as both a comprehension check and as reminders that reinforce the critical aspects of the worker task that are central to our design.

After the 12 redistribution decisions, spectators complete a brief exit survey. The first part of the exit survey consists of three questions. First, we randomly select one of the 12 decisions that the spectators made and present the same information to them. We then ask spectators in the lucky outcomes condition how many encryptions they think workers solved on average. For spectators in the lucky opportunities condition, we randomly draw a multiplier and ask how many encryptions they think workers with that multiplier solved on average. Finally, we ask them how much they would allocate to the winner if they knew they had solved more encryptions.

The second part of the exit survey asks spectators to select their level of agreement with several belief statements in a five-point Likert scale grid. It probes their views on various topics relating to income redistribution and the role of the government. We also embed an attention check in one of the rows that states, "Select disagree if you are reading this."

D. Recruitment of Workers and Spectators

We recruited 2,416 participants on Amazon Mechanical Turk to participate in the worker task in September 2021. We restricted participation to workers who were based in the United States, had a 95% minimum approval rate, and had at least 500 approved human intelligence tasks. We excluded 16 participants who completed fewer than one encryption per minute for a final sample of 2,400 workers. We paid all workers a fixed participation

fee of $2 upon task completion. Workers also received an additional payment of up to $5 based on the decision of a randomly chosen spectator approximately 6 weeks after completing the task.

We recruited 1,170 panelists from the Federal Reserve Bank of New York's *Survey of Consumer Expectations* (SCE) to participate in the spectator task in October and November 2021. This survey targets a nonconvenience, nationally representative panel of US heads of households (Armantier et al. 2017).[13] We conducted two waves of recruitment over 2.5 weeks to reach our target sample size. Our recruitment sample includes all recent panelists who have taken part in SCE over the past 3 years but excludes those who are still actively participating in the SCE core module. We targeted a representative sample of US households based on the same criteria used for entry into the core SCE sample. We invited a total of 3,500 panelists to participate in the experiment over the two waves and had an average response rate of 33.4%.

The experimental interface was mobile-friendly to encourage hard-to-reach demographic groups to participate in our experiment. Given the high-quality nature of the SCE sample, we preregistered to not impose any exclusion criteria for our spectator sample. Our in-sample data-quality metrics support this decision: The median spectator spent 15 minutes on the survey, 89% passed the attention check, and 77% passed all four comprehension questions on their first attempt. No spectator failed to answer more than two comprehension questions. We paid all respondents a $5 Amazon gift card for completing the survey.

Table 1 reports descriptive statistics for the spectator sample.[14] The average spectator is 49 years old; 52% are female, 62% are married, and 14% are nonwhite. More than 62% of spectators attained a college degree, 58% work full-time, 15% work part-time, and 20% are retired. About a quarter (24%) of spectators have a household income below $40,000 per year and about a third (32%) more than $100,000 per year. Our sample includes individuals living in all 50 states plus Washington, DC. About 23% of spectators live in the Midwest, 21% in the Northeast, 35% in the South, and

[13] The SCE recruitment process works as follows. Prospective panelists receive an initial intake survey that is distributed via mail to a random sample of US addresses, which contains a $1 bill and offers an additional $5 payment for completing and returning the paper-based survey that collects detailed demographic information and some initial economic expectations. Respondents are informed that they may be eligible to participate in additional surveys and asked to provide their email address if they would like to participate. Stratified random sampling is then used to selectively invite respondents to join the core SCE panel to achieve a nationally representative sample of US household heads in terms of income, gender, age, race/ethnicity, and census division. Respondents participate in the core SCE survey for up to 12 months and receive a $15 payment for each survey. They can also receive additional payments for completing special surveys, such as the one we fielded for this study. See sec. 3 in Armantier et al. (2017) for additional details about the SCE intake and recruitment process.

[14] See table A2 for summary statistics on our sample of workers. Figure A2*A* plots the distribution of worker performance.

TABLE 1
AVERAGE SPECTATOR CHARACTERISTICS BY TREATMENT CONDITION

		Baseline Condition			Information Treatment		
	All (1)	Lucky Outcomes (2)	Lucky Opportunities (3)	Ex Post Lucky Opportunities (4)	Lucky Outcomes (5)	Lucky Opportunities (6)	Ex Post Lucky Opportunities (7)
				A. Demographic Characteristics and Race			
Age	49.09	49.50	50.06	47.28	48.68	49.63	49.37
Male	.48	.50	.47	.48	.48	.52	.45
Married	.62	.61	.63	.68	.57	.66	.60
Number of children under 18	.60	.51	.57	.61	.53	.74	.62
White	.86	.82	.88	.84	.84	.88	.90
Black	.08	.12	.06	.07	.07	.07	.07
Hispanic	.08	.09	.08	.09	.08	.08	.06
				B. Education and Employment			
Completed college	.62	.62	.62	.65	.60	.67	.58
Numeracy index	4.09	4.00	4.12	4.16	4.11	4.15	4.03
Works full-time	.58	.62	.53	.64	.55	.56	.58
Works part-time	.15	.15	.14	.12	.18	.14	.15
Retired	.20	.16	.22	.19	.19	.23	.22
Homeowner	.72	.68	.77	.70	.69	.74	.77

	C. Household Income						
Income below 40k	.24	.25	.22	.19	.23	.24	.27
Income between 40k and 75k	.28	.30	.27	.28	.28	.25	.26
Income between 75k and 100k	.16	.17	.14	.16	.17	.15	.17
Income over 100k	.32	.25	.36	.38	.31	.36	.28
	D. Region						
Lives in Midwest	.23	.30	.20	.26	.20	.20	.20
Lives in Northeast	.21	.20	.20	.19	.23	.22	.23
Lives in South	.35	.32	.30	.36	.31	.36	.43
Lives in West	.22	.18	.30	.20	.25	.22	.15
Used mobile device	.37	.38	.39	.35	.42	.31	.38
Minutes spent in experiment	14.72	14.67	15.06	13.62	14.83	15.37	15.09
Passed attention check	.89	.86	.85	.92	.93	.91	.88
Observations (spectators)	1,170	197	194	193	197	196	193
Observations (redistributive decisions)	14,040	2,364	2,328	2,316	2,364	2,352	2,316

NOTE.—This table shows the demographic composition of our spectator sample, comparing spectators treated with and without information about the likelihood that the best performer is the winner, between lucky outcomes (cols. 2 and 5), lucky opportunities (cols. 3 and 6), and ex post lucky opportunities (cols. 4 and 7) conditions.

22% in the West. Columns 2–7 show that spectator characteristics are similar across conditions.

IV. Main Results

The primary outcome we examine is the fraction of earnings, r_{ip}, that spectator i redistributes from the winner to the loser in worker pair p. We refer to the worker who initially receives the total earnings as the "winner" and the worker who initially receives no earnings as the "loser." When $r_{ip} = 0$, the loser gets none of the total earnings, and the winner retains all the earnings. Both workers receive half of the total earnings when $r_{ip} = 0.5$. The primary outcomes, hypotheses, and empirical specifications described in this section were preregistered in our preanalysis plan. We note any deviations in the main text.

A. Redistribution under Lucky Opportunities and Lucky Outcomes

Spectators redistribute less income when luck is manifested through unequal opportunities. Panel A of table 2 reports the average level of redistribution across our luck treatments. Under lucky outcomes, spectators redistributed 27.6% of earnings from the winner to the loser on average. In contrast, spectators redistributed only 23.4% of earnings on average under lucky opportunities. Thus, spectators redistributed 4.2 percentage points less of total income when luck was experienced through unequal opportunities ($p < .01$, col. 3). This difference equates to a 15.3% decrease in the final earnings for the losing worker. If we restrict our analysis to observations in which spectators face uncertainty about whether the inequality is merit based, we find that spectators redistributed 5 percentage points ($p < .01$) less of total income under lucky opportunities, implying 17.3% lower earnings for the losing worker.[15]

Next, we examine how redistribution responds to the importance of luck in determining outcomes. Figure 1 plots mean redistribution in each luck treatment against the probability that the best performer lost: $1 - \pi$. Intuitively, $1 - \pi$ can be thought of as the probability that initial earnings inequality was due to luck or, equivalently, the probability that earnings inequality is not merit based. We depict $1 - \pi$ on the x-axis, rather than π,

[15] We refer to situations in which $\pi = 1$ as situations without uncertainty because the information available to spectators in those situations allows them to conclude without a doubt that the winner of a worker pair is the best performer of the pair. For example, this is the case when the coin-flip probability is zero in the lucky outcomes condition or when both workers have the same multiplier in the lucky opportunities condition. Upon excluding these observations, average redistribution equals 28.9% in the lucky outcomes environment versus 23.9% in the lucky opportunities environment.

TABLE 2
FRACTION REDISTRIBUTED AS A FUNCTION OF THE PROBABILITY THAT THE BEST PERFORMER LOST ($1 - \pi$)

	Outcome: Fraction of Earnings Redistributed		
	Lucky Outcomes (1)	Lucky Opportunities (2)	Difference (3)
	A. Average Redistribution		
Constant	.276***	.234***	.042**
	(.010)	(.013)	(.016)
	B. Elasticity of Redistribution		
Probability best performer lost ($1 - \pi$)	.037***	.020***	.017**
	(.005)	(.004)	(.006)
Constant	.331***	.263***	.068***
	(.013)	(.014)	(.019)
	C. Average Redistribution across π-Bins		
$1 - \pi = .0$	.131***	.179***	−.048
	(.018)	(.018)	(.025)
$1 - \pi \in (.0, .05]$	.175***	.198***	−.024
	(.018)	(.016)	(.024)
$1 - \pi \in (.05, .10]$	.202***	.202***	−.000
	(.016)	(.016)	(.022)
$1 - \pi \in (.10, .15]$	.240***	.226***	.014
	(.017)	(.017)	(.023)
$1 - \pi \in (.15, .20]$	.270***	.208***	.062**
	(.017)	(.015)	(.023)
$1 - \pi \in (.20, .25]$	.316***	.226***	.090***
	(.016)	(.015)	(.022)
$1 - \pi \in (.25, .30]$	.345***	.228***	.117***
	(.017)	(.016)	(.023)
$1 - \pi \in (.30, .35]$	.322***	.240***	.082***
	(.016)	(.016)	(.023)
$1 - \pi \in (.35, .40]$	.315***	.249***	.066**
	(.016)	(.016)	(.023)
$1 - \pi \in (.40, .45]$	.336***	.260***	.076**
	(.018)	(.017)	(.025)
$1 - \pi \in (.45, .50)$	.327***	.291***	.037
	(.019)	(.018)	(.026)
$1 - \pi = .50$	.336***	.298***	.038
	(.020)	(.019)	(.028)
Observations (redistributive decisions)	2,364	2,328	4,692

NOTE.—This table shows estimates of redistribution under lucky outcomes (col. 1), lucky opportunities (col. 2), and the difference (col. 3). Panel A shows the mean share of earnings redistributed. Panel B shows the elasticity of the fraction of earnings redistributed with respect to the likelihood that the winning worker performed better than the losing worker. Panel C shows the relationship between redistribution and the likelihood that the winning worker performed better split into 12 bins. Heteroskedasticity-robust standard errors clustered at the spectator level are given in parentheses.

** Significant at the 1% level.

*** Significant at the 0.1% level.

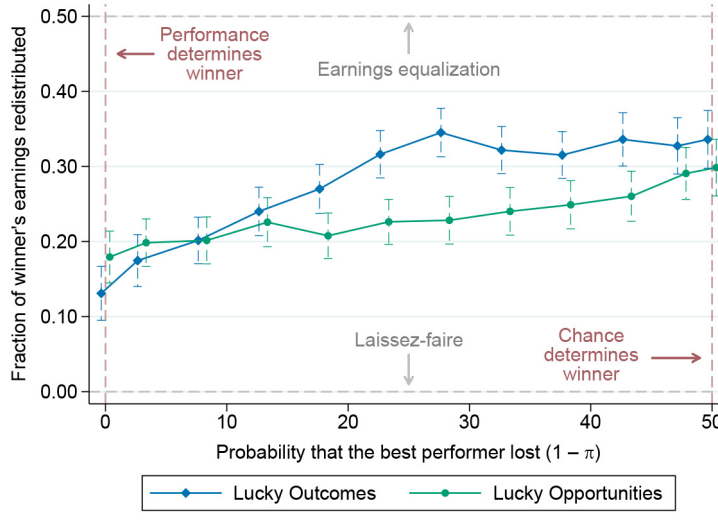


FIG. 1.—Redistribution and probability that best performer lost $(1 - \pi)$. This figure shows the average share of earnings redistributed between workers (from the higher-earning winner to the lower-earning loser) relative to the likelihood that the best performer lost $(1 - \pi)$. The two series represent our main experimental conditions: lucky outcomes (line with diamonds) and lucky opportunities (line with dots). The vertical dashed lines denote 95% confidence intervals.

because this makes it easier to discuss the effect of increasing the impact of luck, which facilitates the exposition of our results. The intercept in figure 1 shows mean redistribution across worker matches when the best performer was always the winner—because workers drew equal multipliers or because there was zero probability of earnings being determined by a coin flip. The slope shows how redistribution changes when there is an increase in the probability that initial earnings inequality was due to luck.

We first examine worker pairs where performance alone determined the winner, eliminating any uncertainty about whether initial earnings inequality stemmed from worker performance or luck. In these merit-based outcomes, spectators behave similarly across luck environments.[16]

[16] Notably, redistribution is slightly higher under lucky opportunities in these merit-based outcomes, contrary to the overall redistribution gap documented in panel A of table 2. This pattern may partly reflect spectator confusion in cases where the winner had the lower multiplier. While winners had higher multipliers in 11 of 12 decisions, for the merit-based outcome case $(1 - \pi_{ip} = 0)$, the winning worker had either a lower or equal multiplier. When the winning worker had the lower multiplier, some spectators may not have noticed this difference from their other decisions. Indeed, when the multipliers were exactly equal, the average redistribution was 10.7%, compared to 19.2% when the winner had a lower multiplier. Although the confidence intervals are relatively large when we analyze these subsamples separately, this suggests that some spectators may have erroneously attributed the lower multiplier to the loser for the pure merit decision.

We next examine how redistribution responds to an increase in the probability that the best performer lost. Under lucky outcomes, spectators' redistribution decisions initially react strongly to this probability. As the probability that the best performer lost increases from 0 to about 25%–30%, the share of total income that spectators redistribute increases from 13.1% to 34.5%. Beyond that point, redistribution does not respond to further increases in the probability that the best performer lost. In contrast, spectators' reactions are more muted when there are lucky opportunities. As the probability that the best performer lost increases from 0 to the range 25%–30%, average redistribution increases from 17.9% to 22.8%. However, spectators continue to respond to the increased role that luck plays in determining outcomes beyond that point, increasing the amount they redistribute in an approximately linear manner.

To summarize the relationship between average redistribution and the likelihood that the best performer in match p lost in each luck environment, we estimate linear models that relate the share of earnings redistributed to $1 - \pi_{ip}$:

$$r_{ip} = \alpha + \beta(1 - \pi_{ip}) + \varepsilon_{ip}, \tag{7}$$

where ε_{ip} is an error term. The main parameter of interest is $\beta = \partial \mathbb{E}(r_{ip})/\partial \mathbb{E}(1 - \pi_{ip})$, which measures the elasticity of redistribution with respect to $1 - \pi_{ip}$. The exogenous within-subject variation in π_{ip} allows us to identify β.

Spectators respond less to changes in the probability that the outcome was luck based when luck stems from unequal opportunities instead of coin flips. Panel B of table 2 presents estimates of β across the different luck environments. A 10 percentage point increase in $1 - \pi_{ip}$ leads to a $\hat{\beta} = 3.7$ percentage point increase in the share of earnings redistributed in the lucky outcomes condition (col. 1). In contrast, redistribution is less elastic to changes in $1 - \pi_{ip}$ in the lucky opportunities condition: A 10 percentage point increase in $1 - \pi_{ip}$ leads to a $\hat{\beta} = 2.0$ percentage point increase in redistribution when luck emerges through unequal opportunities (col. 2). The difference between estimated elasticities is statistically significant ($p < .01$, col. 3).[17]

This differential sensitivity to changes in $1 - \pi_{ip}$ leads to a substantial redistribution gap across our experimental treatments. Column 3 of panel C in table 2 presents the average difference in redistribution for each

[17] These results remain robust when we exclude the two extreme cases $1 - \pi_{ip} = 0$ and $1 - \pi_{ip} = 0.50$. When excluding all observations that correspond to these extreme cases from the analysis, we find that a 10 percentage point increase in $1 - \pi_{ip}$ leads to a 3.5 percentage point increase in redistribution under lucky outcomes and a 1.8 percentage point increase under lucky opportunities, maintaining the 1.7 percentage point gap in responsiveness between the two environments.

π-bin, along with the estimated standard error.[18] We observe a significant difference in redistribution for 6 out of the 10 π-bins in which spectators face inferential challenges when trying to determine whether the inequality was merit based ($p < .05$).[19]

B. Margins of Redistribution and Heterogeneity

Differences in redistribution across luck environments can be decomposed into differences in the extensive margin (whether anything is redistributed) and the intensive margin (how much is redistributed).[20] In appendix section B.4, we investigate the role of the extensive and intensive margins in accounting for the redistribution gap across luck environments. On the extensive margin, we find that a higher fraction of spectators do not redistribute any earnings under lucky opportunities than under lucky outcomes ($p < .01$). On the intensive margin, we still find that redistribution is less responsive to the likelihood that the best performer lost under lucky opportunities than lucky outcomes ($p < .01$). Thus, both the extensive and the intensive margin contribute to the redistribution gap.

We also find that redistribution behavior varies by respondents' gender, income, and political views. Table A3 presents the results from a heterogeneity analysis. Females tend to favor redistribution more than males across both luck environments. Respondents in households with annual incomes below $100,000 redistribute more on average across both luck environments. This finding is notable in light of recent cross-country evidence by Maréchal et al. (2025) suggesting that actual redistribution policies are more strongly correlated with the preferences of lower socioeconomic groups. Finally, Republicans tend to support redistribution less than others. These patterns are consistent with previous evidence and

[18] Formally, in panel C of table 2, we estimate regressions of the form

$$r_{ib} = \sum_{b=1}^{12} \gamma_b (1 - \pi_b) + \varepsilon_{ib},$$

where $1 - \pi_b$ is an indicator that equals 1 if $1 - \pi_{ip}$ is in bin b. We estimate this equation separately for each treatment and interact the bins with treatment dummies to assess formally whether mean redistribution is the same across luck treatments at a given π-bin. We cluster standard errors at the spectator level in all specifications.

[19] When making decisions for 2 of the 12 π-bins (i.e., in the two extreme cases: $1 - \pi_{ip} = 0$ and $1 - \pi_{ip} = 0.50$), spectators trying to assess whether merit could have determined the winner do not face the same inferential challenges as in the other 10 decisions. This is because $1 - \pi_{ip} = 0$ corresponds to cases in which the probability of a coin flip is zero (in lucky outcomes) or the winner does not have any relative advantage (in lucky opportunities) so that the winner is for sure the better performer. Similarly, $1 - \pi_{ip} = 0.50$ corresponds to cases in which winning is independent of performance because the coin-flip probability is 100% (in lucky outcomes) or the winning worker's relative advantage was too large to be overcome (in lucky opportunities).

[20] This analysis of the intensive and extensive margins of redistribution was not included in our preregistration and should therefore be considered exploratory in nature.

extend to the domain of unequal opportunities (Alesina and Giuliano 2011; Ashok, Kuziemko, and Washington 2015; Alesina, Stantcheva, and Teso 2018; Almås, Cappelen, and Tungodden 2020; Cohn et al. 2023).

C. *The Impact of Observable Features on Redistribution*

To understand why spectators are less responsive to changes in the probability of the best performer losing under lucky opportunities compared to lucky outcomes, we conduct a two-step analysis. First, we examine how spectators incorporate observable features of the luck environment into their redistribution decisions (sec. IV.C.1). Second, we analyze the relationship between these features and the probability that the best performer lost (sec. IV.C.2).

1. Spectator Responses to Unequal Opportunities

We begin by analyzing how spectators incorporate unequal opportunities into their redistribution decisions. We create two intuitive measures of unequal opportunities: (1) the arithmetic difference between the productivity multipliers of the winner and loser ($m_1 - m_2$), and (2) the ratio of these multipliers (m_1/m_2). We then evaluate which measure better predicts spectators' redistribution choices by regressing the fraction of earnings redistributed on each measure (table 3). We find that when either the multiplier difference or ratio is included in the regression by itself, each has a significant impact on redistribution (cols. 1 and 2). However, when both measures are included simultaneously in the regression model, only the multiplier difference continues to have a significant effect (col. 3).[21]

These results suggest that spectators primarily rely on the multiplier difference (and not the ratio) when making redistribution decisions in the lucky opportunities environment. For example, spectators treat multipliers $(m_1, m_2) = (2, 1)$ the same as $(m_1, m_2) = (3, 2)$ when making redistribution decisions, when in fact it is more likely that outcomes were luck based in the former; specifically, $1 - \pi(2/1) = 0.46$ compared to $1 - \pi(3/2) = 0.38$. A worker competing against another with a multiplier of 1 would only require a multiplier of 1.5 to have an equivalent advantage to a worker with a multiplier of 3 competing against one with a multiplier of 2.

Figure 2 provides visual evidence of how spectators use the features from the environment in each luck condition to inform their redistribution decisions. The figure presents a series of binned scatterplots plotting

[21] The coefficient on the multiplier difference in col. 3 is not statistically different from the corresponding estimate in col. 1 ($p = .42$). However, we can reject that the coefficient on the multiplier ratio is the same in cols. 2 and 3 ($p < .01$).

TABLE 3
IMPACT OF UNEQUAL OPPORTUNITIES ON REDISTRIBUTION DECISIONS

	Outcome: Fraction of Earnings Redistributed		
	(1)	(2)	(3)
Multiplier difference	.041***		.051***
	(.007)		(.015)
Multiplier ratio		.042***	−.012
		(.008)	(.015)
Constant	.202***	.171***	.213***
	(.014)	(.017)	(.019)
Observations (redistributive decisions)	2,328	2,328	2,328

NOTE.—This table shows the average redistribution (from the winner's earnings to the loser) as a function of the difference between the multiplier of the winner and the loser ($m_1 - m_2$) and the ratio between the multiplier of the winner and the multiplier of the loser (m_1/m_2) in the lucky opportunities environment. Heteroskedasticity-robust standard errors clustered at the spectator level are given in parentheses.

*** Significant at the 0.1% level.

average redistribution (*y*-axis) against the information observed by spectators (*x*-axis). Under lucky outcomes (fig. 2*A*), the relationship between redistribution and the coin-flip probability is piecewise linear. For low coin-flip probabilities, there is a positive and linear relationship, but for sufficiently large coin-flip probabilities, redistribution is unresponsive to further changes in the coin-flip probability. Under lucky opportunities (fig. 2*B*), the relationship between redistribution and the multiplier difference is linear across the entire range of multiplier differences.[22] On average, a one-unit increase in the difference between the high and low performers' multipliers leads to a 4 percentage point increase in the share of earnings redistributed.

2. Features of the Environment and Source of Inequality

Next, we study the empirical relationship between the observable features of each luck environment and the probability that the inequality between workers is the result of luck. Under lucky outcomes, there is a linear relationship between the coin-flip probability and the likelihood that the best performer lost (fig. 3*A*). This relationship starts at the origin, as the best performer never loses when the coin-flip probability is zero, and increases linearly with the coin-flip probability.

In contrast, under lucky opportunities, there is a concave relationship between the multiplier difference and the likelihood that the best performer

[22] In table A10, we regress the fraction of earnings redistributed on higher-order polynomials of the multiplier difference and find that none of these terms have a significant effect on redistribution. This provides further evidence suggesting that spectators use the linear multiplier difference to inform their redistribution decisions.

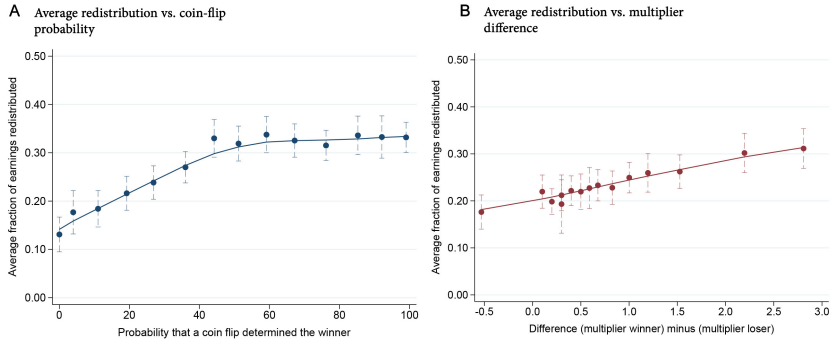


FIG. 2.—Relationship between redistribution and features of luck environment observed by spectators. This figure shows the average earnings redistributed from the winner to the loser (*y*-axis) as a function of the probability that a coin flip determined the winner (*A*) or the difference between the productivity multiplier of the winner and the loser (*B*), both divided into 15 equal-sized bins. Solid lines plot the predicted values from local polynomial estimates. Dashed vertical lines show 95% confidence intervals.

lost (fig. 3*B*). When multipliers are equal, the best performer never loses. However, even a small increase in the multiplier difference leads to a steep rise in the probability of the best performer losing. For example, increasing the multiplier difference from 0 to 0.5 increases the likelihood of the best performer losing from 0% to about 20%. As the multiplier difference becomes sufficiently large, the best performer lost whenever they did not receive the high multiplier.

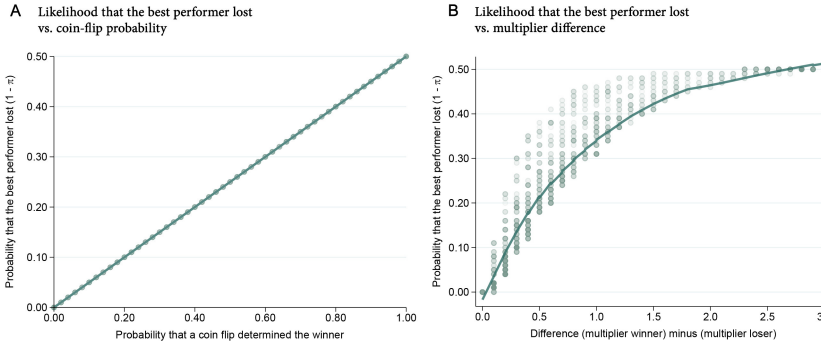


FIG. 3.—Relationship between probability that best performer lost and features of luck environment observed by spectators. This figure shows the probability that the best performer lost (*y*-axis) as a function of the probability that a coin flip determined the winner (*A*) or the difference between the productivity multiplier of the winner and the loser (*B*). The dots denote the empirical frequencies in our experimental data, and the intensity of the shading indicates the number of observations at a given point. The solid lines depict the best fit using either linear specification (*A*) or lowess regression (*B*).

Taken together, the combination of how redistribution responds to the environmental features and how these features relate to the probability that the best performer lost explains why spectators underreact under unequal opportunities. Under lucky outcomes, the coin-flip probability and the likelihood that the best performer lost are linearly related (fig. 3*A*). As a result, the relationship between redistribution and the coin-flip probability (fig. 2*A*) exactly mirrors the relationship between redistribution and the probability that the best performer lost (fig. 1, line with diamonds).

In the lucky opportunities condition, spectators' redistribution decisions respond linearly to the multiplier difference across its entire range (fig. 2*B*). However, this linear response fails to account for the concave relationship between the multiplier difference and the probability that the best performer lost (fig. 3*B*). As a result, spectators appear to "underreact" (redistribute less than expected based on the actual likelihood that the best performer lost) for relatively small values of $1 - \pi$ and "overreact" for larger values (fig. 1, line with dots).[23] In section V, we investigate whether this linear response to multiplier differences persists when we also provide direct information about π.

V. The Role of Information and Effort Responses

We explore two broad categories of mechanisms that may drive the patterns of redistribution that we observe across luck environments. First, we investigate the role of inaccurate beliefs about the probability the best performer lost through an information intervention. Second, we examine the role of actual or perceived differences in worker performance across luck environments in explaining our results.

A. Information Intervention

A key question is why spectators are less responsive to changes in the likelihood that the winner was luck based in lucky opportunities versus lucky outcomes, which leads to substantial gaps in average redistribution across these environments. To examine whether this is due to spectators facing inferential challenges when assessing the impact of unequal opportunities on earnings inequality, we implement an information treatment. In this treatment, in addition to the information spectators receive in the baseline

[23] In app. sec. B.3, we show that the same relationship prevails even if lucky opportunities are not multiplicative but additive. More precisely, we consider the possibility that workers receive unequal, additive boosters, capturing the idea that a relative advantage can resemble a head start sometimes. In such an environment, the probability that the inequality is merit based is still convex in the relative advantage (the difference in boosters), implying that $1 - \pi$ is concave in the booster difference—as is the case for our design using productivity multipliers.

condition, we provide spectators with the information they would otherwise need to infer. Specifically, for each of their 12 redistribution decisions, we tell spectators the likelihood that the best performer won, π.

We begin by examining how the information intervention affects the relationship between redistribution and the likelihood that the outcome is luck based. If inferential challenges are the primary driver of the linear redistribution pattern, then we expect the treatment to significantly alter this relationship, as spectators no longer need to rely on heuristics or potentially inaccurate inference. In contrast, if the linear pattern persists, this would suggest that other factors shape redistributive behavior. Then, we analyze how the information intervention affects spectators' reliance on observable features of the luck environment.

Providing readily accessible information about π made redistribution more responsive to the likelihood that the best performer lost in both luck environments. Figure 4 plots mean redistribution in the baseline and information interventions. In the baseline lucky outcomes condition, a 10 percentage point increase in the likelihood that the best performer lost raised redistribution by 3.7 percentage points. In the information-treated group, this effect increases by 1.5 percentage points ($p < .05$, table A4, panel B) to 5.2 percentage points in total. Similarly, in the baseline lucky

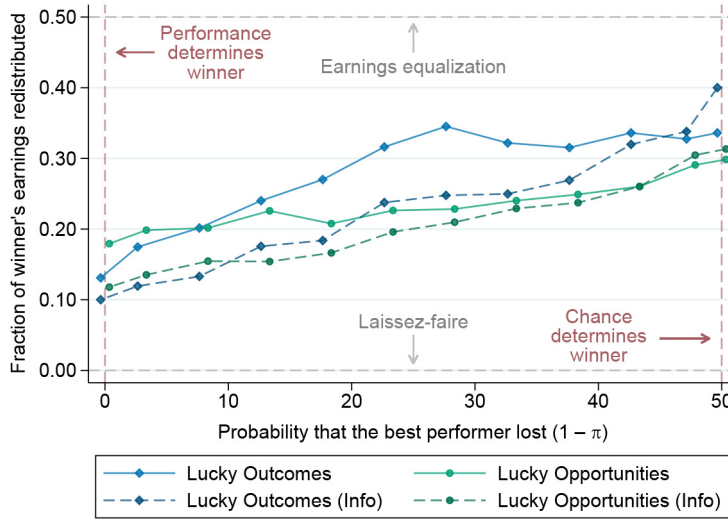


FIG. 4.—Effect of providing information about π. This figure shows the average share of earnings redistributed between workers (from the higher-earning winner to the lower-earning loser) as a function of the likelihood that the best performer lost for our two luck environments with and without information about π. The solid lines depict our baseline lucky outcomes (diamonds) and lucky opportunities environments (circles). The dashed lines depict redistribution for the lucky outcomes (diamonds) and lucky opportunities (circles) environments with information about π.

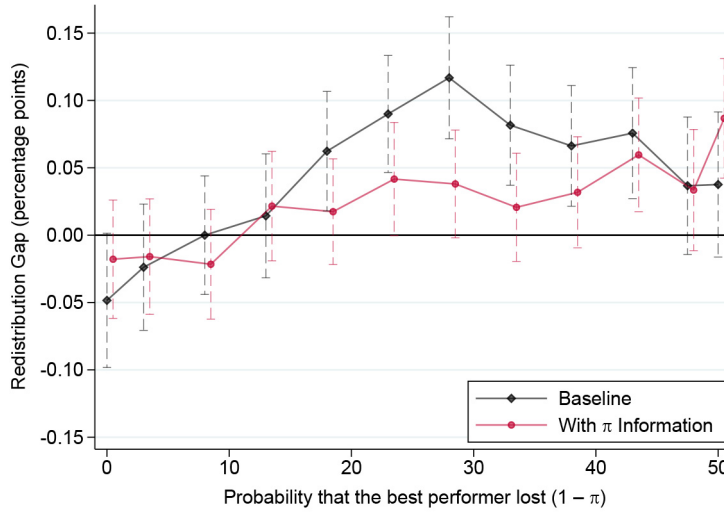


FIG. 5.—Redistribution gap by $1 - \pi$ in baseline and information treatments. This figure shows the gap in the average share of earnings redistributed between workers (from the higher-earning winner to the lower-earning loser) between the lucky outcomes and lucky opportunities environments as a function of the likelihood that the best performer lost. The line with diamonds depicts the baseline treatments in which spectators do not observe π. The line with dots depicts the information treatment in which spectators have direct access to the exact value of π when making their redistribution decisions.

opportunities condition, a 10 percentage point increase in the likelihood that the best performer lost raised redistribution by 2.0 percentage points. In the information-treated group, this effect increases by 1.7 percentage points ($p < .01$) to 3.7 percentage points in total, implying an 85% relative increase in the elasticity of redistribution. In other words, when information about the probability that earnings inequality was due to luck becomes readily accessible, spectators become more sensitive to this probability and adjust their redistribution decisions accordingly.[24]

As a consequence of this increased sensitivity, the information treatment reduced the redistribution gap across luck environments. Figure 5 plots the redistribution gap between the lucky outcomes and lucky opportunities environment in the baseline treatments with no information about π (line with diamonds) and readily accessible information about π (line with

[24] The effects of providing information about π are robust to again excluding the two extreme situations $1 - \pi_{ip} = 0$ and $1 - \pi_{ip} = 0.50$ where spectators face less inferential challenges. Excluding these observations from the analysis of the information treatments, we find that a 10 percentage point increase in the likelihood that the best performer lost raises redistribution by 4.8 percentage points under lucky outcomes (versus 3.5 percentage points in the baseline condition) and by 3.6 percentage points under lucky opportunities (versus 1.8 percentage points in the baseline condition).

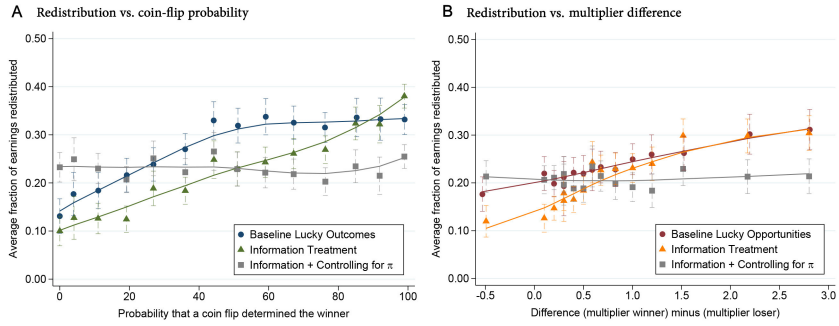


FIG. 6.—Relationship between redistribution and observable luck features (information intervention). *A*, Average share of winners' earnings redistributed by probability that coin flip determined outcomes. *B*, Average share redistributed by gap in worker multipliers. Information-treated spectators are provided information about π in addition to the observable features of the luck environment.

dots). The redistribution gap diminishes substantially when information about π is provided, with the most pronounced declines observed over the range $1 - \pi \in [20, 40)$. A joint F-test of the null hypothesis that the change in the redistribution gap between lucky opportunities and lucky outcomes across all π-bins equals zero is rejected at the 1% significance level ($F = 2.19$, $p = .01$). Notably, the gap is small and statistically indistinguishable from zero for 10 of the 12 π-bins, and is statistically significant at $p < .05$ for only two π-bins (table A5, col. 2). This finding is consistent with the interpretation that spectators face inferential challenges when making redistribution decisions.

Next, we investigate how the information intervention affected the spectators' reliance on the other observable features of the two luck environments.[25] Figure 6 shows a series of binned scatterplots analogous to those in figure 2. Circle markers reproduce the relationship between mean redistribution and the observable features in the baseline luck environments. Triangle markers show this same relationship but under information treatments. Finally, square markers show the relationship between mean redistribution and the observable information, after removing the influence of π on redistribution.[26] Solid lines plot the predicted values from local polynomial estimates.

[25] We did not preregister this analysis; however, our linearity results from sec. IV.C.2 invite further exploration.

[26] More precisely, we plot the residual of an individual's redistribution after subtracting from it the component explained by the likelihood that the best performer lost. To construct these residuals, we first regress the share of earnings redistributed on the likelihood that the best performer lost (as in eq. [7]) and estimate the residuals from this regression, adding back the unconditional sample mean to facilitate the interpretation of units.

The information treatment altered how spectators incorporated observable features of the luck environment into their redistribution decisions. Figure 6 illustrates how the relationship between redistribution and coin-flip probabilities/multiplier differences changed once information on the likelihood that inequality is merit based became readily available to spectators. In the lucky outcomes environment, the information treatment had two effects: It decreased mean redistribution for most coin-flip probabilities, and it increased the responsiveness of redistribution for relatively high coin-flip probabilities. In the lucky opportunities environment, the information treatment reduced mean redistribution for relatively low multiplier differences. Moreover, it made redistribution more responsive to changes in the multiplier difference for relatively small multipliers, particularly in the range where a small increase in relative multipliers has a substantial impact on worker outcomes.

The relationship between mean redistribution and the observable features of the luck environment appears to be mediated by the likelihood that the outcome was merit based. Controlling for the likelihood that the outcome is merit based attenuates the relationship between redistribution and the features of the environment observed by spectators (fig. 6, gray lines). Once this likelihood is controlled for, these features have no independent impact on redistribution.[27] This finding further supports the hypothesis that spectators primarily use observable features of the environment as proxies to infer the likelihood of merit-based outcomes.

B. Timing of Luck and Worker Performance

An important difference between our luck environments is that lucky outcomes occur after completing the task, while unequal opportunities are known before. This aspect of the lucky opportunities condition reflects how luck typically arises in many real-life situations. This difference in the timing of luck could drive the differences in the redistribution decisions that we observe if spectators have different expectations about how workers respond to getting a high or low multiplier.[28]

[27] Table A11 provides the corresponding regression estimates and confirms that the multiplier difference has no significant effect on redistribution when controlling for π in the information intervention.

[28] What matters for redistribution behavior are spectators' beliefs about worker performance, which we control for using our additional treatments. Empirically, we do not find any evidence that worker performance responds to receiving a high or low multiplier (see table A7). We also observe no differences in the overall performance distribution across luck environments (see fig. A3). A Kolmogorov–Smirnov test for equality of distribution cannot reject the hypothesis that the distribution of worker performance in the lucky outcomes and lucky opportunities environments is equal ($p = .909$). Notably, other work has found performance responses to differences in opportunities (Andre 2025). This difference likely arises because our environment is a winner-take-all tournament with a fixed working period, while Andre (2025) considers differential piece-rate wages and allows

To examine whether the timing of lucky opportunities affects redistribution, we implemented an additional between-subjects treatment that aligned the timing of when workers learn about their luck across environments. In the *ex post lucky opportunities* condition, workers learn their multipliers only after they complete the task. This is in contrast to our baseline lucky opportunities condition, in which we inform workers of their multipliers before they begin working on the encryption task. However, this timing is in line with when workers learn about the probability that a coin flip determines their earnings in the lucky outcomes environment, namely after completing the task. To ensure this intervention is known to spectators, we provide them a visual timeline of the worker task sequence and require that they pass a comprehension check about whether multipliers are revealed to workers before or after the task.[29]

We find that whether opportunities are encountered before or after working has no economically or statistically significant impact on spectators' redistribution decisions. In table A8, we reestimate the primary specifications in table 2 but compare redistribution between the baseline and ex post lucky opportunities. We find no significant differences in the average redistribution: The average amount of income redistributed was 23.4% in baseline lucky opportunities versus 24.4% in ex post lucky opportunities ($p = .57$). We also find no significant differences in the elasticity of redistribution to changes in luck ($p = .89$). Figure 7 plots our estimates of the average redistribution for both lucky opportunities conditions across each π-bin. Across the entire range of π-bins, we find no differences in the level of redistribution.[30]

To further assess whether spectators expected differential effort from workers who received a low versus high multiplier, we elicited their stated beliefs about worker performance across the multiplier distribution. To do this we randomly selected a multiplier for each spectator in the baseline lucky opportunities condition and elicited their beliefs about the average number of encryptions completed by workers who received that multiplier. This information enables us to recover the perceived elasticity of worker performance with respect to the productivity multiplier, which

workers to choose how long they work. Indeed, DellaVigna et al. (2022) find that higher incentives lead to higher output when workers can choose how long they work but have no effect when there is a fixed working period.

[29] The equivalent visual timelines and comprehension checks are provided to all spectator conditions, and not only to those spectators in ex post lucky opportunities. See figs. 5 and 18 in the supplementary materials (https://joyzwu.github.io/papers/prsw_supplementary.pdf).

[30] Accordingly, there is still a redistribution gap as well as a difference in the elasticity of redistribution between the ex post lucky opportunities condition and the lucky outcomes condition. For example, if the likelihood that inequality is due to luck increases by 10 percentage points, redistribution increases by 1.6 percentage points ($p < .01$) less under ex post lucky opportunities (versus 1.7 percentage points in our main comparison in sec. IV.A).

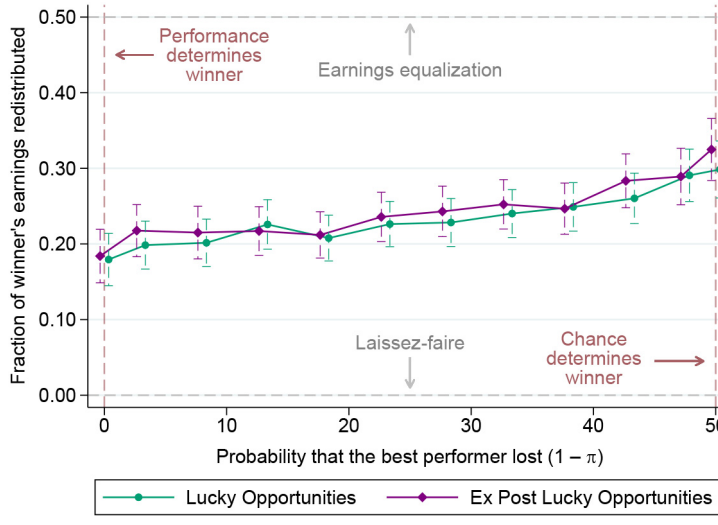


FIG. 7.—Redistribution by $1 - \pi$ in baseline and ex post lucky opportunities conditions. This figure shows the average share of earnings redistributed between workers (from the higher-earning winner to the lower-earning loser) as a function of the likelihood that the best performer lost. It depicts two variations of the lucky opportunities condition: Workers in the baseline condition are aware of their multiplier prior to beginning the encryption task, and workers in the ex post condition learn their multiplier after completing the encryption task.

we estimate by regressing spectator beliefs about worker performance on the randomly selected multiplier (table A9). We find no evidence that spectators expect a significant worker performance response from receiving a high or low multiplier.[31]

Finally, it is possible that spectators believed that merely knowing there would be coin flips or multipliers causes workers to put in differential effort across our luck environments, even if the exact coin-flip probability or productivity multiplier is revealed to workers only after completing the task. To fully eliminate any scope for differential beliefs about workers' performance responses, we implemented additional subtreatments in which workers did not receive any information about how the winner would be determined prior to working on the task. Crucially, spectators in this "rules-after" scenario in both the lucky outcomes and ex post lucky opportunities conditions knew that workers faced identical instructions before completing the task. We analyze the data from these subtreatments in appendix section B.5. In figure B3, we show that whether workers learn

[31] Spectators could also expect that the difference in performance between the winner and loser is not identical across our luck environments. In app. sec. B.6, we show that the effort gap between winners and losers is very similar in the two luck conditions across the entire range of π-bins. Moreover, we provide evidence that potential differences in the winner-loser effort gap cannot explain the redistribution gap between the two luck environments.

about the possibility of a coin flip or multipliers affecting the outcome before or after working on the task has no impact on spectators' support for redistribution. In addition, figure B4 and the estimates in table B3 show that there is still a redistribution gap as well as a difference in the elasticity of redistribution between lucky outcomes and lucky opportunities in the rules-after treatment, that is, when workers had identical information before completing the task.

VI. Predicting Political and Social Views

We document that individuals redistribute less when luck arises through unequal opportunities rather than exogenous sources. But which environment is more predictive of the spectator's political and social views? Our data allow us to examine the external relevance of the experimental measure of redistribution attitudes based on our lucky opportunities environment and compare it to measures based on the lucky outcomes environment used in prior work. To assess this, we estimate the correlation between the average earnings redistributed by spectators in each environment and 14 self-reported political and social views.[32]

Redistribution behavior under lucky opportunities correlates more with real-world social and political attitudes than the lucky outcomes environment. We find that mean redistribution in the lucky opportunities condition is more predictive than in lucky outcomes for 13 out of the 14 social and political attitudes we measure (table A6). To create summary measures of political attitudes, we calculate a *z*-score that averages the 14 self-reported individual attitudes and an index that equals the first component of a principal component analysis. The summary indexes reveal that behavior in the lucky opportunities condition is 70% more predictive of attitudes than in the lucky outcomes condition. While the overall explanatory power is modest, with R^2-values ranging from 1.4% to 4.3%, the lucky opportunities environment consistently explains nearly 3 times as much variation in political attitudes as the lucky outcomes environment. This finding suggests that redistribution decisions made in the lab better reflect real-world social and political views when luck arises through unequal opportunities rather than unequal outcomes.

VII. Discussion

Meritocratic fairness ideals contend that individuals are willing to tolerate inequalities due to differences in effort but oppose those arising from

[32] We collected participants' social and political views to conduct validation of our experimental measures and heterogeneity analysis but did not preregister specific hypotheses. As a result, we consider the analysis in this section exploratory.

chance. But when people face unequal opportunities, this distinction is obfuscated by the fact that luck and effort are intertwined. As a result, individuals may find it difficult to assess the source of inequality and may misattribute success to effort rather than luck. This paper shows that impartial third parties are more tolerant of inequality when luck arises through unequal opportunities, and there is uncertainty about the role of luck in determining outcomes.

We provide evidence that a key mechanism driving these results is the differential ability to understand the impact of luck across environments, as spectators face greater inferential challenges in mapping features of the luck environment to their consequences for outcomes when luck manifests through unequal opportunities.

Our results offer a potential rationale for the apparent disconnect between the previous experimental literature and observed patterns of US inequality. Research that generates inequality through exogenous variation in outcomes has found that most Americans equalize incomes when income differences are due to luck (Almås, Cappelen, and Tungodden 2020). However, support for redistribution in the United States remained stagnant over a period when differences in opportunities became increasingly important (Chetty et al. 2014; Ashok, Kuziemko, and Washington 2015). Consistent with these trends, we show that redistribution is less sensitive to changes in luck when luck arises through unequal opportunities. Similarly, the United States remains the most unequal country in the OECD while ranking poorly on equality of opportunity (Corak 2013; Mitnik, Helsø, and Bryant 2022). Consistent with these cross-country comparisons, we show that Americans tolerate more inequality when it arises due to differential opportunities.

We conclude by discussing two implications of our results for models that seek to understand and predict attitudes toward redistribution. First, we document that in the absence of readily available information about the likelihood that outcomes are merit based, spectators rely on simple heuristics when factoring the impact of luck into their redistribution decisions. As a result, people fail to appreciate how small differences in initial circumstances can greatly impact outcomes. Providing information about the importance of luck reduces this reliance on heuristics. Models that seek to accommodate cognitive errors hold some promise for predicting and explaining how beliefs shape redistribution attitudes.

Second, our results show that readily available information about the likelihood that outcomes are merit based greatly impacts people's redistribution decisions. This suggests that the information individuals frequently encounter might disproportionately impact their views on inequality and redistribution. For example, popular media coverage (e.g., rags-to-riches stories) may lead individuals to have a greater tolerance for inequality of opportunities, making them less willing to correct this source of unfairness

through redistribution. Exploring how salient information shapes individuals' tolerance for inequality is a promising avenue for future research.

Taken together, our results highlight that redistributive behavior is not invariant to how luck is implemented in the lab. The lucky opportunities environment has several important features that affect redistribution, which the simpler lucky outcomes paradigm cannot capture. We provide a portable, tractable, and rich environment to study income redistribution when there are unequal opportunities that can inform the development of inequality models and the design of optimal redistribution policies.

Data Availability

Data and code replicating the tables and figures in this article can be found in Preuss et al. (2025) in the Harvard Dataverse, https://doi.org/10.7910/DVN/H7QOXG.

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